Library / Toolkit	Category	JavaScript / TypeScript Frameworks			Notes
		React	Svelte	Vue	
3D CityDB WebMap	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Moderate	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Web-based viewer for semantic 3D city models using CesiumJS https://www.3dcitydb.org/3dcitydb-web-map/1.8.0/3dwebclient/index.html
AntV L7	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install @antv/l7 @antv/larkmap	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Heatmaps, 3D layers, vector rendering https://antv.antgroup.com/en - github.com/antvis/L7
Bertin.js	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): Moderate	☑ npm install bertin d3	☑ npm install bertin d3	☑ npm install bertin d3	Thematic mapping engine for cartograms, flow maps, and choropleths https://www.3dcitydb.net/3dcitydb-web-map/1.8.4/3dwebclient/splash/SplashWindow_Mobile.html
CARTO.js	3D Rendering: Moderate Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install @carto/react-core @carto/react-api @carto/react-ui	✕ Not Supported	✕ Not Supported	Location intelligence toolkit with spatial filtering, SQL, and map styling. https://github.com/CartoDB/carto.js
CesiumJS	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install cesium & npm install resium	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Planetary-scale 3D visualization; ideal for terrain & satellites https://cesium.com/platform/cesiumjs/
D3.js	3D Rendering: Simulated Web Mapping: Yes Web Graphics Library (WebGL): Partial Big Data Visualization (Vz): Yes	☑ npm install d3	☑ npm install d3	☑ npm install d3	GeoJSON rendering, Projection transforms, custom map styling https://d3js.org/
deck.gl	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install deck.gl @deck.gl/react	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	GPU-accelerated layers; great for large-scale data rendering www.github.com/visgl/deck.gl
Flatgeobuf.js	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install flatgeobuf	☑ npm install flatgeobuf	☑ npm install flatgeobuf	High-performance binary format for spatial data; supports streaming and indexing https://flatgeobuf.org/
GeoJSON-VT	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install geojson-vt	☑ npm install geojson-vt	☑ npm install geojson-vt	Converts GeoJSON into vector tiles for fast rendering in Mapbox GL or Leaflet https://github.com/mapbox/geojson-vt/blob/main/README.md
Geokdbush	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install geokdbush kdbush	☑ npm install geokdbush kdbush	☑ npm install geokdbush kdbush	Fast spatial indexing using K-D trees https://github.com/mourner/geokdbush/releases
Globe.GL	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install globe.gl three	☑ npm install globe.gl three	☑ npm install globe.gl three	UI component for GlobUI component for Globe Data Visualization using ThreeJS/WebGLe Data Visualization using ThreeJS/WebGL https://globe.gl/
Google Maps JS API	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install @vis.gl/react-google-maps	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Full API support; autocomplete, directions, etc. https://developers.google.com/maps/documentation/javascript/overview
GeoRasterLayer	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): Moderate	☑ npm install georaster georaster-layer-for-leaflet	☑ npm install georaster georaster-layer-for-leaflet	☑ npm install georaster georaster-layer-for-leaflet	Leaflet plugin for rendering raster data from GeoTIFFs https://www.npmjs.com/package/georaster-layer-for-leaflet
Geotiff.js	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install geotiff	☑ npm install geotiff	☑ npm install geotiff	Parses and renders GeoTIFF raster data in-browser; supports elevation and imagery https://github.com/geotiffjs/geotiff.js/
HERE Maps JS API	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Full API support; autocomplete, directions, etc. https://www.here.com/developer/javascript-api
JSTS	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): Yes	☑ npm install jsts	☑ npm install jsts	☑ npm install jsts	JavaScript Topology Suite - port of JTS for spatial predicates and geometry operations https://github.com/bjornharrtell/jsts
Kepler.gl	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install --save @kepler.gl/components & others	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	WebGL-powered, supports CSV/GeoJSON ingestion https://kepler.gl/
Leaflet	3D Rendering: Simulated Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): Moderate	☑ npm install leaflet react-leaflet	☑ npm install svelte-mapbox mapbox-gl	☑ npm install vue-mapbox-gl mapbox-gl	Lightweight, tile-based; great for OSM and simple overlays https://leafletjs.com/
Mapbox GL JS	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install react-map-gl mapbox-gl	☑ npm install svelte-mapbox mapbox-gl	☑ npm install vue-map-gl mapbox-gl	Rich styling, camera control, vector overlays https://docs.mapbox.com/mapbox-gl-js/api/
MapLibre GL JS	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install react-map-gl maplibre-gl	☑ npm install react-map-gl maplibre-gl	☑ npm install @indoorequal/vue-maplibre-gl maplibre-gl	Open-source fork of Mapbox GL JS; no vendor lock-in https://maplibre.org/maplibre-gl-js/docs/
Mapzen.js	3D Rendering: Limited Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Moderate	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Lightweight wrapper for Mapzen vector tiles and Tangram rendering https://github.com/expandaventures/mapzen.js
Mobility.js	3D Rendering: No Web Mapping: No Web Graphics Library (WebGL): No Big Data Visualization (Vz): No	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Lightweight toolkit for visualizing movement data (trajectories, stops, flows) https://github.com/cblanquera/mobility
OpenLayers	3D Rendering: Limited Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): Moderate	☑ npm install ol	☑ npm install ol	☑ npm install ol vue3-openlayers	Advanced CRS, WMS/WFS, and spatial logic https://openlayers.org/
Placemark	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): No	☑ npm install placemark	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	Web-based editor; useful for curriculum and POI authoring https://testdev.tools/placemark/
Proj4js	3D Rendering: No Web Mapping: Moderate Web Graphics Library (WebGL): No Big Data Visualization (Vz): No	☑ npm install proj4	☑ npm install proj4	☑ npm install proj4	Coordinate system conversion; supports custom CRS https://proj4js.org/
Rbush	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install rbush	☑ npm install rbush	☑ npm install rbush	2D spatial indexing with fast bounding box queries; rbush-knn adds nearest-neighbor search https://github.com/mourner/rbush
react-geo	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Moderate	☑ npm install @terrestris/react-geo	✕ Not Supported	✕ Not Supported	Rich UI components for legends, layer toggles, and map controls https://terrestris.github.io/react-geo/
react-globe.gl	3D Rendering: Yes Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Yes	☑ npm install react-globe.gl	☑ npm install react-globe.gl	☑ npm install react-globe.gl	A React component to represent data visualization layers on a 3-dimensional globe in a spherical projection https://www.npmjs.com/package/react-globe.gl
react-spatial	3D Rendering: Limited Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Moderate	☑ npm install react-spatial	✕ Not Supported	✕ Not Supported	Mobility tracking, live overlays, and routing https://react-spatial.geops.io/
Shapefile-js	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Moderate	☑ npm install shapefile.js	☑ npm install shapefile.js	☑ npm install shapefile.js	Reads ESRI Shapefiles in-browser and converts to GeoJSON https://github.com/matthewdowns/shapefile.js/releases
Supercluster	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install supercluster	☑ npm install supercluster	☑ npm install supercluster	Fast clustering engine for millions of points — used in Mapbox and MapLibre https://github.com/mapbox/supercluster
Tangram	3D Rendering: Moderate Web Mapping: Yes Web Graphics Library (WebGL): Yes Big Data Visualization (Vz): Limited	☑ npm install leaflet tangram	⚠ Manual integration or community wrapper	⚠ Manual integration or community wrapper	browser-based mapping engine designed for real-time rendering of 2D and 3D maps https://tangrams.readthedocs.io/en/main/
TopoJSON	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): Indirect Big Data Visualization (Vz): Yes	☑ npm install topojson-client	☑ npm install topojson-client	☑ npm install topojson-client	Extension of GeoJSON that encodes topology — ideal for compact thematic maps https://github.com/mapbox/supercluster
Turf.js	3D Rendering: No Web Mapping: Yes Web Graphics Library (WebGL): No Big Data Visualization (Vz): Limited	☑ npm install @turf/turf	☑ npm install @turf/turf	☑ npm install @turf/turf	Buffer, intersect, distance, etc. https://www.turfjs.org/